# SUPPLEMENTARY MATERIAL FOR "ESCALATION OR SHARED EPISODES? A JOINT SPATIO-TEMPORAL MODEL OF ARMED CONFLICT AND SOCIAL UNREST"

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This supplement contains the detailed tables and figures referred to in the main paper. Equation, section and table numbers without the prefix S refer to the main paper.

## S1. Data.

S1.1. *Construction of the cell-week table.* The grid consists of all $0.5^\circ$ cells that intersect a country polygon of Natural Earth (1:10m), excluding cells without a terrestrial biome (rock and ice, and large lakes). Each cell is assigned to the country that contains its centroid. A coastal cell whose centroid falls in the sea is assigned to the nearest country. Every cell-week appears exactly once, and national indicators are attached through the assigned country, with the exceptions described below. The result is 59,339 cells observed for up to 366 weeks.

The table built by the original pipeline differs in one important respect. About 8.5% of its 34.7 million distinct cell-weeks (before the biome exclusion) occur more than once, with multiplicities such as $2^k$, $3^k$ and $2 \cdot 3^k$, the signature of many-to-many joins between the grid and polygons of countries, IPC regions and national indicators. These copies make up 50.5% of its 70.0 million rows. The copies of a cell-week share its events but often disagree on the covariates. Among duplicated cell-weeks, the copies differ in the EPI and the liberal democracy index in 62% of cases, in GDP in 49%, in the IPC phases in about 25% and in the climate variables in 7–11%. We do not know why the climate values of a cell differ between copies; the original pipeline does not record which polygon each copy comes from. We keep the whole first row of each cell-week, so that all covariates of a row come from the same copy. The climate values of duplicated cell-weeks are therefore those of an arbitrary copy, and this part of the error is retained rather than repaired. The climate covariates are controls in the joint models and the conclusions do not rest on their coefficients. IPC phase populations are the exception: they are averaged over the copies. Five national indicators (EPI, liberal democracy index, greenhouse gas emissions, material and ecological footprint) are re-attached for each country and year from the cells of that country that were not duplicated. When a country-year has no value in these cells, which happens when the indicator is missing for that country, the code falls back to the most common value of the kept first copies of that country-year. Since every country has non-duplicated cells, such values come from copies joined to a neighbouring country's polygon. The fallback therefore can give a country its neighbour's value and turns a missing value into an observed one, so the missingness indicator is not set. In the whole repaired table, 14.5% of rows take at least one national indicator from this fallback. Access to electricity is not re-attached at all: every cell-week keeps the value of its first copy, which in cells that had copies can come from a neighbouring country. It is therefore not constant within a country-year, and its decomposition into between- and within-country parts mixes differences across borders with change over time. GDP per capita is a national quantity: World Bank GDP of the country divided by the sum of the GHSL population of its cells, and it is constant within each country-year. Section S4.1 describes the consequences of the duplication for the original analysis.

Two features of ACLED matter for modelling. First, geographic coverage expanded over time, and several countries entered the dataset after 2018. Cell-weeks in a country before its coverage start are *unobserved*, not event free, so they are removed rather than coded as zeros. Coverage start dates are taken from ACLED's country coverage table. Territories not listed there take the date of the country or region that contains them. Before the biome exclusion, coverage removes 7,078,822 cell-weeks in 79 countries and territories; 3,998,319 of them are in Antarctica, which the biome exclusion would remove in any case. The event history is treated as incomplete in the first four covered weeks. After both exclusions the table has 19,364,639 cell-weeks. Second, the probability that an event is recorded depends on media and partner density, so the outcome combines occurrence and reporting (Weidmann, 2016). The two are not separately identified without auxiliary information. The latent fields of Section 3 absorb spatially smooth reporting differences together with unmeasured drivers of risk, and we interpret them accordingly. Reporting intensity also changes over time and by country, for example when ACLED adds sources or partners. The year effects are global, and at test time they are held at their 2022 value, so they cannot absorb country-specific changes in reporting. Such changes may explain part of the rise in recorded unrest above, and they can inflate both the estimated state dependence and the persistence of the space-time fields.

ACLED also codes the spatial and temporal precision of each event. An event whose exact location is unknown is placed at the centroid of the town or administrative unit named in the source, and an event whose exact date is unknown at a nominal date within the reported period. The data available to us are weekly counts by cell, and they carry no precision codes, so we cannot filter or weight events by precision. Imprecisely located events accumulate in the cells that contain administrative centroids and capitals. This can inflate both the persistence of events in those cells and the cell-level latent effects estimated there, and it is one more reason to read those effects as recorded rather than actual risk.

S1.2. *Covariates and missing values.* The climate variables enter all models only through their means over the four weeks before the current week, so that they precede the outcome; values of the current week are not used. The original analysis also used the current week's values. A raw temperature means different things in different climates. In the latent models, the five climate variables (maximum and minimum temperature, precipitation, drought and $PM_{2.5}$) therefore enter as standardised anomalies with respect to a cell- and calendar-month-specific climatology estimated from the training years 2018–2022,

$$z_{st} = \big(x_{st} - \bar{x}_{s,\mathrm{m}(t)}\big)/\mathrm{sd}_{s,\mathrm{m}(t)},$$

where $\mathrm{m}(t)$ is the calendar month of week $t$. Precipitation is transformed to $\log(1+x)$ before the anomaly is computed. In dry seasons of arid cells the training-period standard deviation can be zero or close to zero, and any later rain would then give an extreme anomaly. We therefore floor $\mathrm{sd}_{s,\mathrm{m}}$ at the 10th percentile of the positive cell-month standard deviations of the same variable, and set $z_{st} = 0$ where the climatological standard deviation is zero. In the AME region the floor applies, by construction, to 10% of the 164,628 cell-months for temperature and precipitation. The drought index is constant over the training years in 138,600 cell-months (84%), so its anomaly is zero there, and the floor applies to a further 1.3%. The drought anomaly is therefore informative only in a minority of cell-months. The $PM_{2.5}$ anomaly is worse. Its climatological standard deviation is undefined in 91,392 cell-months (56%), where $PM_{2.5}$ is missing, and the floor computed from the other cell-months is about $10^{-9}$, so it does not prevent extreme anomalies. The $PM_{2.5}$ anomaly is missing in 55.5% of training rows, is then set to zero and has no missingness indicator. We do not interpret its coefficient. The anomaly coefficients are identified from variation over time within cells. They are therefore not confounded by the time-invariant spatial pattern of the climatology.

TABLE S1
*Covariates, resolution and source.*

| Covariate | Resolution | Source |
|---|---|---|
| Minimum / maximum temperature | cell–week | CHIRTS |
| Heat index, wet-bulb globe temperature | cell–week | CHIRTS |
| Precipitation | cell–week | CHIRPS |
| $PM_{2.5}$ | cell–week | CAMS |
| Drought index | cell–week | Global Drought Observatory |
| Events of each type in previous 4 weeks | cell–week | ACLED |
| Population | cell | GHSL |
| River length, lake area | cell | Natural Earth |
| Malaria risk | cell | Malaria Atlas Project |
| Biome | cell | RESOLVE Ecoregions |
| Food insecurity (IPC phases 1–5) | subnational–period | IPC |
| GDP per capita | country–year | World Bank |
| Access to electricity | country–year | World Bank |
| Environmental Performance Index | country–year | Yale EPI |
| Liberal democracy index | country–year | V-Dem |
| GHG emissions, material and ecological footprint p.c. | country–year | OWID, GFN |

They can still be confounded by regional shocks that vary over time, since the year effects are global. The cell's mean over the training years enters separately as a cell-level covariate. A five-year climatology is noisy, especially for precipitation, and a longer reference period would be preferable where available. The anomalies are therefore best read as departures from recent conditions.

*Missing covariates.*. Table S1 hides substantial missingness. In the repaired table over the AME region (4,976,634 cell-weeks), the IPC phase populations are missing in 62.7% of rows, $PM_{2.5}$ in 55.5%, and GDP in 3.6%, mainly in South Sudan, Yemen, Somaliland, Eritrea, Western Sahara and Syria. The other national indicators (EPI, liberal democracy index, greenhouse gas emissions, material and ecological footprint) are missing in 0.3% of rows, at most about 850 cell-weeks in any country. These shares are after the fallback described in Section 2.1, which fills some missing values with a neighbouring country's value. The climate means over the previous four weeks are missing in 1.1% of rows, and the event histories in 1.1%. Over all cell-weeks worldwide the IPC phases are missing in 89.6% of rows and precipitation in 20.8%. IPC analyses are carried out mainly where food crises are expected, and several of the countries without GDP data are among the most conflict-affected in the region. Missingness is therefore unlikely to be at random with respect to the outcome.

Apart from that fallback, we keep missing values as missing through the construction of the features. In particular, a cell-week without an IPC analysis has a missing IPC value, not a food-insecure population of zero. All covariates of the latent models other than the event histories are standardised with training-period means and standard deviations. Missing values are then set to zero, the training mean, and a missingness indicator enters the model for each covariate group: one for each national indicator and one for IPC (Section 3.1).

## S2. Further methods.

S2.1. *Finite-volume discretisation and sea barriers.* applying the divergence theorem with a two-point flux approximation gives

$$\tau\Big\{\kappa^2 a_i w_i + \sum_{j\sim i} \frac{\ell_{ij}}{d_{ij}}(w_i - w_j)\Big\} = \mathcal{W}(A_i) \sim \mathcal{N}(0, a_i),$$

where $a_i$ is the area of cell $i$, $j \sim i$ ranges over the cells sharing an edge with $i$, $\ell_{ij}$ is the length of the shared edge and $d_{ij}$ the great-circle distance between the centres. For a grid with spacings $\Delta\lambda, \Delta\varphi$ and centre latitude $\varphi_i$,

$$a_i = R^2 \Delta\lambda \{\sin(\varphi_i + \Delta\varphi/2) - \sin(\varphi_i - \Delta\varphi/2)\},$$
$$\ell_{ij}/d_{ij} = \Delta\varphi/(\cos\varphi_i \, \Delta\lambda) \quad \text{(east–west neighbours)},$$
$$\ell_{ij}/d_{ij} = \cos\varphi_{ij} \, \Delta\lambda/\Delta\varphi \quad \text{(north–south neighbours)},$$

with $\varphi_{ij}$ the latitude of the shared edge. Writing $\boldsymbol{C} = \mathrm{diag}(a_i)$ and $\boldsymbol{G}$ for the weighted graph Laplacian with edge weights $\ell_{ij}/d_{ij}$, we get $\tau \boldsymbol{K} \boldsymbol{w} = \boldsymbol{C}^{1/2} \boldsymbol{\epsilon}$ with $\boldsymbol{K} = \kappa^2 \boldsymbol{C} + \boldsymbol{G}$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, so

(S2.1)

$$\boldsymbol{w} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}_s^{-1}\}, \qquad \boldsymbol{Q}_s(\rho, \sigma) = \tau^2 \boldsymbol{K} \boldsymbol{C}^{-1} \boldsymbol{K}, \qquad \log|\boldsymbol{Q}_s| = 2n \log\tau + 2\log|\boldsymbol{K}| - \textstyle\sum_i \log a_i.$$

$\boldsymbol{Q}_s$ has at most 13 non-zeros per row. The construction is the lumped-mass finite-volume analogue of the finite-element representation in Lindgren, Rue and Lindström (2011). Because it uses the physical cell geometry, it accounts for the shrinking east–west spacing towards the poles. It can also be restricted to land cells by deleting rows and columns, and it wraps around in longitude for global analyses. The continuum relations between $(\kappa, \tau)$ and $(\rho, \sigma)$ hold on the lattice only when $\rho$ spans several cells. A fitted range below two or three cells (100–150 km) indicates a field that the $0.5^\circ$ lattice does not resolve; its $\hat{\sigma}$ is then not the marginal standard deviation, and we interpret such fields as cell-level effects.

REMARK 1. The discretisation imposes reflecting (Neumann) conditions on the boundary of the lattice, which roughly doubles the marginal variance of boundary cells (Section S3.2; see Khristenko et al., 2019, for the effect of boundary conditions on SPDE fields). For a land grid the boundaries are the coastlines. Extending the lattice into the sea removes the inflation but lets correlation leak across narrow seas, for example between Yemen and Somalia across the Gulf of Aden, which is substantively wrong for conflict risk. We therefore use the barrier model of Bakka et al. (2019): the SPDE is written as

$$u(s) - \nabla \cdot \left\{ \tfrac{r(s)^2}{8} \nabla u(s) \right\} = r(s) \sqrt{\pi/2} \, \sigma \, \mathcal{W}(s),$$

which reduces to (3.2) with $\kappa = \sqrt{8}/\rho$ when $r(s) \equiv \rho$. The lattice is extended by six rings of cells ($3^\circ$) without data. Sea cells, identified from the Natural Earth country polygons, get range $r(s) = f\rho$. Land cells without data, such as those outside a regional analysis box, keep the range $\rho$, so that the field is not artificially decorrelated at the edge of the region. The finite-volume form is $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{D}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \boldsymbol{C} + (\rho^2/8)\boldsymbol{G}_m$, where $\boldsymbol{G}_m$ has edge weights $2m_i m_j/(m_i + m_j) \cdot \ell_{ij}/d_{ij}$, $m_i = r_i^2/\rho^2$ (the harmonic mean, the usual two-point flux choice for a discontinuous coefficient), and $\boldsymbol{D} = \mathrm{diag}\{r_i \sigma (\pi a_i/2)^{1/2}\}$, so that $\boldsymbol{Q}_s = \boldsymbol{K}\boldsymbol{D}^{-2}\boldsymbol{K}$. Table S5 reports the marginal variance at the coast and the correlation across straits of two to six cells for several choices of $f$, at $\rho = 400$ km. At this range, $f = 0.1$, the value of Bakka et al. (2019), gives a barrier range of 40 km, less than one cell, and the coastline behaves almost like a reflecting boundary. We use $f = 0.5$. At $\rho = 400$ km it keeps the coastal variance within about 30% of nominal and the correlation across straits of four cells or more (about 220 km) below 0.04, while a strait of two cells still has a correlation of 0.16. Straits narrower than one cell are not resolved at this grid resolution. These figures depend on $\rho$, because the barrier range $f\rho$ and the buffer of six rings scale differently with it. The fitted ranges of the static fields in the joint models are 450–640 km for conflict and 360–430 km for unrest, and those of the space-time fields 470–840 km. Below 400 km the barrier range $f\rho$ is shorter, and the coastline is closer to a reflecting boundary than in Table S5. For ranges of 800 km

or more, $f\rho$ exceeds the buffer, so the coastal variance inflation returns and correlation leaks across the wider straits. Table S5 is also computed for a sea strait inside a closed rectangle of land, not for the geometry of the application, which has an open coast, a buffer of six rings and a reflecting outer edge. It shows the order of magnitude of the effects, not their size in the application. The stored real-data fits use the sea cells of the country polygons as barriers.

S2.2. *Lattices and boundaries of the space-time field.* $v(s,t) = v_{s,\lceil t/\Delta \rceil}$. We use $\Delta = 1$ and $\Delta = 4$ in the simulations, and $\Delta = 13$ (West Africa and the Sahel) and $\Delta = 26$ (Africa and the Middle East) for the joint models of Section 3.4. In the application, $v$ is also coarsened in space: it is defined on the $1^\circ$ cells that contain observed $0.5^\circ$ cells, and each $0.5^\circ$ cell takes the value of its $1^\circ$ cell. The field $v$ is defined on the observed land cells only, with no buffer, so it cannot link cells across the sea. Its lattice has reflecting boundaries at the coast and at the edges of the analysis box. This departs from Remark 1, under which land cells outside a regional box keep the range $\rho$. Near these edges the prior variance of $v$ is inflated, roughly doubled at an edge. West Africa and the Sahel is a box of $34^\circ \times 21^\circ$, about $3700 \times 2300$ km, and the fitted ranges of the space-time fields there are 420–790 km, so a substantial part of the box lies within one range of an edge. Estimates of $\delta$ and $\kappa_{\mathrm{U}\to\mathrm{C}}$ conditional on $v$ should be read with this caveat. In the joint models, the static field $w$ uses the same lattice as elsewhere: the observed cells, a buffer of six rings and sea barriers with $f = 0.5$ (Remark 1). The prior on $\operatorname{atanh}\phi$ is $\mathcal{N}(0, 1.5^2)$. Non-separable alternatives exist (Lindgren et al., 2024) but are not needed for the questions asked here.

S2.3. *Joint model: identification and related models.*

*The copy construction..* A loading in the design matrix would make $\boldsymbol{A}$ depend on $\boldsymbol{\theta}$. We instead keep $\boldsymbol{A}$ fixed and put $\lambda$ in the prior precision, as in the copy construction of R-INLA. Let $\boldsymbol{a}$ be the vector of the $m$ values of $v$ and introduce a second vector

$$\boldsymbol{b} = \lambda \boldsymbol{a} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \tau_c^{-1}\boldsymbol{I}), \qquad \tau_c = 10^4.$$

Conflict rows of the design load on $\boldsymbol{a}$ and unrest rows on $\boldsymbol{b}$. The standard deviation of $\boldsymbol{\varepsilon}$ is $0.01$, small compared with that of $v$, so the unrest predictor receives $\lambda v$ up to a negligible perturbation. If $\boldsymbol{Q}_a = \boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$ is the precision of $\boldsymbol{a}$, the precision of $(\boldsymbol{a}, \boldsymbol{b})$ is

$$\boldsymbol{Q}_{ab} = \begin{pmatrix} \boldsymbol{Q}_a + \lambda^2\tau_c\boldsymbol{I} & -\lambda\tau_c\boldsymbol{I} \\ -\lambda\tau_c\boldsymbol{I} & \tau_c\boldsymbol{I} \end{pmatrix} = \boldsymbol{R}_{ab}^\top \boldsymbol{R}_{ab}, \qquad \boldsymbol{R}_{ab} = \begin{pmatrix} \boldsymbol{R}_a & \boldsymbol{0} \\ -\lambda\tau_c^{1/2}\boldsymbol{I} & \tau_c^{1/2}\boldsymbol{I} \end{pmatrix},$$

with $\boldsymbol{R}_a^\top \boldsymbol{R}_a = \boldsymbol{Q}_a$, and $\log|\boldsymbol{Q}_{ab}| = \log|\boldsymbol{Q}_a| + m\log\tau_c$. The loading therefore does not enter the log-determinant, and the explicit factor $\boldsymbol{R}_{ab}$ is used for posterior sampling (Section 3.5). The sparsity pattern of $\boldsymbol{Q}_{ab}$ does not depend on $\lambda$. The loading has a $\mathcal{N}(0, 2^2)$ prior. The loading of conflict is fixed at one, which fixes the scale and the sign of $v$, so $\lambda$ and its sign are identified. With $\lambda = 0$, the conflict equation of (3.4) is the univariate conflict model with its own space-time field, and the two outcomes are independent given their histories. The comparison of the joint fit with that univariate fit shows the effect of letting unrest inform $v$, under the assumption that there is no outcome-specific space-time variation (see *Identification and limits* below). In the application $v$ uses $\Delta = 13$ weeks on the $1^\circ$ lattice of Section 3.3.

The shared component mixes common shocks to occurrence with common shocks to reporting. Reporting rates vary in space (Weidmann, 2016), reporting of protest clusters in space and time through sequential attention (Hellmeier, Weidmann and Rød, 2018), and non-fatal events are under-reported selectively (Croicu and Eck, 2022). A period of attention to a place raises the recorded counts of both event types, and the joint model treats it as a shared episode. The two sources of co-movement could be separated with several independent sources of event data (Cook et al., 2017). With a single source they cannot. We therefore

read $\kappa_{\mathrm{U}\to\mathrm{C}}$ as conditional on shared episodes of either kind, and we do not interpret $v$ as occurrence or as reporting. The loading $\lambda$ also mixes the two. It reflects how strongly unrest follows the shared episodes and how differently the two event types are reported.

Model (3.4) is a linear model of coregionalisation with a single space-time component and fixed loading one for conflict (Gelfand et al., 2004). Outcome-specific persistent variation enters through $u_j$, $e_j$ and $w_j$, but there is no outcome-specific space-time field. Space-time variation of conflict that is not shared with unrest therefore also loads on $v$, and $\lambda$ is estimated against it. Space-time variation of unrest that is not shared with conflict loads on $v$ as well, with the attenuation of $\hat{\kappa}_{\mathrm{U}\to\mathrm{C}}$ described above. A second space-time component, as in the extended model, relaxes this at the cost of a second latent block of the size of $v$. Multivariate Hawkes models of event data (Jun and Cook, 2024) and discrete-time Hawkes models of ACLED (Browning et al., 2026) explain co-occurrence by triggering, with a background rate that does not contain a shared time-varying latent component. In such models a shared episode can only appear as cross-excitation. Model (3.4) can be read as adding a shared stochastic background to cross-lagged terms. Spatio-temporal SPDE models of political violence (Python et al., 2019; Egbon et al., 2024) have so far modelled one outcome at a time.

S2.4. *Computation.* with step length $\ell$ halved until the objective does not decrease. If $\ell$ falls below $10^{-6}$, the step is taken even if the objective has decreased. For the joint model (3.4) the rows of both outcomes are stacked, so $\boldsymbol{y}$, $\boldsymbol{\pi}$ and $\boldsymbol{W}$ run over the cell-weeks of both. The Laplace approximation to the marginal likelihood is

The hyperparameters are set to $\hat{\boldsymbol{\theta}} = \arg\max\{\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\}$ by L-BFGS-B on the internal scale: logarithms of ranges and standard deviations, $\operatorname{atanh}\phi$, and the loading $\lambda$ itself. The internal parameters are bounded: standard deviations to $[0.005, 20]$, ranges to $[10, 2\times10^4]$ km, $\operatorname{atanh}\phi$ to $[-3, 3]$ and $\lambda$ to $[-5, 5]$. The bounds keep the inner problem well conditioned; an estimate at a bound would indicate a poorly identified hyperparameter. Gradients are forward differences with step $10^{-4}$ on the internal scale, taken inwards at an upper bound. Each evaluation warm-starts Newton at the previous mode. Given $\hat{\boldsymbol{\theta}}$, the latent posterior is approximated

version of INLA (Rue, Martino and Chopin, 2009) and is similar in spirit to TMB (Kristensen et al., 2016). Laplace approximations are generally accurate for binary GLMMs but can understate uncertainty when there are few observations per latent variable and events are rare (Fong, Rue and Wakefield, 2010; Ferkingstad and Rue, 2015). The intervals also ignore the uncertainty in $\hat{\boldsymbol{\theta}}$. Section S3.2 therefore checks coverage at the event prevalence of the data, and Section S3.3 checks it for the space-time model, where each latent value is informed by only $\Delta$ Bernoulli observations. When a space-time field is fitted and the data contain one, the plug-in standard errors are too small. In S1 the ratio of the empirical standard deviation of $\hat{\delta}$ to its mean standard error is between 1.19 and 1.66, and the coverage of nominal 95% intervals is between 0.80 and 0.95. In S2 without escalation the ratio for $\hat{\kappa}_{\mathrm{U}\to\mathrm{C}}$ is 1.20 (Section 4.1). Intervals from the space-time and joint models in the application are therefore likely too narrow, and an estimate whose nominal interval only just excludes zero should not be read as clearly different from zero.

*Computation.*. All prior precisions are sparse. Every Newton step and every log-determinant requires a factorisation of $\boldsymbol{Q} + \boldsymbol{H}$, whose sparsity pattern does not depend on $\boldsymbol{\theta}$. The design $\boldsymbol{A}$ has a few dense columns, those of the fixed effects, and many sparse ones. Writing $\boldsymbol{A} = (\boldsymbol{X}_d, \boldsymbol{S})$, we store $\boldsymbol{X}_d$ as a dense array and assemble $\boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W} \boldsymbol{A}$ by blocks. The blocks $\boldsymbol{X}_d^\top \boldsymbol{W} \boldsymbol{X}_d$ and $\boldsymbol{S}^\top \boldsymbol{W} \boldsymbol{X}_d$ are accumulated over chunks of 250,000 rows by dense matrix products (BLAS), which bounds the memory used by each process. Only $\boldsymbol{S}^\top \boldsymbol{W} \boldsymbol{S}$ is

computed as a sparse product. The dense blocks are stored with their explicit zeros, so the sparsity pattern of $\boldsymbol{Q} + \boldsymbol{H}$ is the same at every evaluation.

Two factorisations are implemented. The first is the supernodal Cholesky factorisation of CHOLMOD, through scikit-sparse. Its symbolic analysis, which includes the fill-reducing ordering, is computed once per model and reused for every numerical factorisation. The second, used when CHOLMOD is not available, is a sparse LU factorisation in symmetric mode (SuperLU). For matrices of dimension above 2,000 its fill-reducing ordering is a nested dissection computed once by METIS (Karypis and Kumar, 1998). If METIS is not available, the code falls back to multiple minimum degree on $\boldsymbol{M}^\top + \boldsymbol{M}$, with $\boldsymbol{M} = \boldsymbol{Q} + \boldsymbol{H}$. For the space-time model the nested-dissection ordering reduced factorisation time by a factor of 7 to 14 compared with that fallback. An LU factorisation stores both triangular factors, so it needs about twice the memory of a Cholesky factor of the same matrix. scikit-sparse was not installed on the computing cluster, so all fits run there, including every joint fit reported in the main text, used SuperLU.

The $n_\theta$ perturbed evaluations of a finite-difference gradient are independent. They run in parallel in forked worker processes, which share the data, the design and the symbolic analysis without copying them and start Newton from the current mode. The joint model (3.4) has $n_\theta = 12$ hyperparameters: two for the country effects, two for the cell effects, four for the static fields and four for the shared field. Newton iterations stop when the relative change of the objective is below $10^{-8}$ and the largest step is below $10^{-5}$, or after 50 steps. An inner solve that stops at 50 steps without converging gives a Laplace value that is meaningless and often spuriously high. Such evaluations are rejected, as are those at which the factorisation fails: the objective is set to a very large value. In a parallel gradient, a component whose perturbed evaluation is rejected is set to zero. If the evaluation at the current point is itself rejected, the components are not set to zero and take very large values. In the serial path, the finite differences of L-BFGS-B are taken on the large value directly. Rejected evaluations can make the optimiser stop at a poor point. We did not record their number for the real-data fits; as a check, warm and cold starts of the extended joint model in West Africa reached the same optimum (Section 3.5), and the subsampled and full fits agreed closely.

Since $\boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W} \boldsymbol{A}$ aggregates over weeks, the cost of the spatial blocks is driven by the number of cells, not by the number of cell-weeks. A global $0.5^\circ$ land grid gives about $6 \times 10^4$ latent values per spatial field. The space-time block has $nT_v$ latent values, and twice as many in the joint model, which is why it is fitted on regional subsets or on a coarser lattice. The cost of forming $\boldsymbol{H}$ grows with the number of cell-weeks and the number of fixed effects. For the joint model both are doubled, since each outcome has its own rows and its own fixed effects.

S2.5. *State dependence versus persistent heterogeneity.* Recent events in the same cell are the strongest single predictor of events (Weidmann and Ward, 2010; Mueller and Rauh, 2022; Menicali, Marcantonio and Castruccio, 2026). This has several explanations, which model (3.1) separates only partially:

1. *state dependence*: recorded events raise the probability of further recorded events in the same place, through retaliation, mobilisation or the arrival of armed groups;
2. *persistent heterogeneity*: some places are simply riskier ($u$, $w$, $e$ or $v$ is large there), and past events are informative only as a proxy for that latent risk (Heckman, 1981);
3. *reporting dynamics*: media attention and reporting networks themselves persist, so a place that has been reported on is more likely to be reported on again, whatever happens there (Weidmann, 2016).

All three produce strong autocorrelation in $Y_{st}$ and a large predictive contribution of $h_{st}$. In the conflict literature the distinction appears as the question of whether apparent contagion

between neighbouring places reflects spillover or shared conditions (Buhaug and Gleditsch, 2008). Self-exciting point process models address it by decomposing intensity into a background and a triggering component (Mohler et al., 2011; Mohler, 2013; Porter and White, 2012; Clark and Dixon, 2018; Reinhart, 2018; Jun and Cook, 2024).

In model (3.1), $\delta$ is the effect of recent recorded history conditional on the latent persistence $u + w + e + v$, and it is identified only under the assumed latent structure. It is conditional on heterogeneity at the scales that are modelled. Persistent heterogeneity that none of the latent terms can represent loads onto $\delta h_{st}$ and inflates $\hat{\delta}$. This is why the cell effect $e$ is included: without it, time-invariant risk that varies between adjacent cells could only be represented by $h_{st}$. The space-time field $v$ is resolved at steps of $\Delta$ weeks, and $h_{st}$ covers the previous four weeks. When the two time scales are similar, separating $\delta$ from $v$ relies on the AR(1) form, so we report $\delta$ for several $\Delta$ (Section S3.3). We therefore read $\hat{\delta}$ as conditional dependence of recorded events, not as a causal contagion effect. To limit the initial-conditions problem of dynamic random-effects models (Wooldridge, 2005), we drop weeks whose four-week history is incomplete (the first weeks of each cell and of each country's ACLED coverage). We also add each cell's history at its first usable week as a covariate, following Wooldridge (2005). This is a limited version of that device. A single four-week window is a noisy proxy for a cell's initial state, and its coefficient does not vary in time, whereas under the AR(1) form of $v$ the dependence of later weeks on the initial state decays with time. The device therefore only partly removes the initial-conditions bias.

## S3. Further simulation studies.

S3.1. *Data-generating process.* The simulator mimics the structure of the ACLED grid. The domain is a $40 \times 30$ block of $0.5^\circ$ cells between $10^\circ$E and $30^\circ$E and between $5^\circ$S and $10^\circ$N (1,200 cells of about 55 km), with 104 training weeks, 52 test weeks and 12 burn-in weeks. Countries are the Voronoi cells of randomly placed seeds (14 countries). Four biomes are latitude bands with boundaries perturbed by a smooth random field, ordered from "cold" (biome 1) to "tropical" (biome 4). Population is a standardised Matérn field (range 250 km) plus white noise. Temperature and precipitation anomalies are standardised AR(1) processes in time (coefficients 0.6 and 0.3) of Matérn fields with ranges of 800 and 500 km. A raw temperature is also generated, as a climatology $30 - 1.2(\varphi - \varphi_{\min})$ in latitude $\varphi$, which rises from biome 1 to biome 4, plus a seasonal cycle $(1 + 0.15|\varphi|)\cos(2\pi t/52)$ and 2.5 times the anomaly. The climatology and the seasonal cycle have no effect in the truth. Three national covariates $Z_0, Z_1, Z_2$ are standard normal across countries. $Z_1$ and $Z_2$ have no effect. Outcomes are generated sequentially from

(S3.1)
$$\operatorname{logit} \pi_{st} = \beta_0 + \beta_{\text{pop}} \text{pop}_s + \delta \log\left(1 + \textstyle\sum_{l=1}^{4} y_{s,t-l}\right) + \beta(s)\, \text{temp}_{st} + \beta_{\text{prec}} \text{prec}_{st} + \alpha Z_{0,c(s)} + u_{c(s)} + w(s) + v(s,t),$$

with $\beta(s) = \gamma_{b(s)} + \eta(s)$ and $u_c \sim \mathcal{N}(0, 0.3^2)$. To avoid generating data from the model used for fitting, $w$, $\eta$ and the innovations of $v$ are drawn from Gaussian processes with the continuous Matérn covariance ($\nu = 1$) in great-circle distance, evaluated at the cell centres. They are not drawn from the lattice GMRF of Section 3.2. Unless stated otherwise, $\beta_0 = -4$, $\beta_{\text{pop}} = 0.5$, $\delta = 0.6$, $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$, $\eta \equiv 0$, $\beta_{\text{prec}} = -0.15$, $\alpha = 0$, $\sigma_w = 1$, $\rho_w = 400$ km and $v \equiv 0$. The resulting prevalence of event weeks is about 4%.

The default parameters are not calibrated to the application, and all covariate effects in (S3.1) are linear on the logit scale.

With $R$ replicates, a proportion $p$ (coverage, rejection rate) has Monte Carlo standard error $\{p(1-p)/R\}^{1/2}$, at most 0.09 for $R = 30$ and 0.11 for $R = 20$, and a mean estimate has Monte Carlo standard error equal to its empirical SD divided by $R^{1/2}$. We give these standard errors where a comparison depends on them.

TABLE S2

*S0: accuracy of the finite-volume SPDE field and of the empirical Bayes Laplace approximation, with fields drawn from the continuous Matérn process (50 replicates per setting). Mean over replicates, with the SD across replicates in parentheses, except for the coverage of $b_1$, where it is the Monte Carlo standard error.*

| Quantity | common ($b_0 = -2$, 50 reps) | rare ($b_0 = -4.5$, 50 reps) |
|---|---|---|
| Prevalence of event weeks | 0.161 (0.030) | 0.019 (0.005) |
| $\hat{\rho}_w$ (km) [truth 400] | 420 (40) | 409 (51) |
| $\hat{\sigma}_w$ [truth 1] | 0.884 (0.064) | 0.875 (0.075) |
| $\hat{b}_1$ [truth 0.5] | 0.494 (0.016) | 0.502 (0.021) |
| Coverage of 95% interval for $b_1$ | 0.88 (0.05) | 0.96 (0.03) |
| Coverage of 95% intervals for $b_0 + w(s)$ (share of cells) | 0.948 (0.009) | 0.943 (0.017) |
| $\mathrm{cor}(\hat{w}, w)$ | 0.954 (0.008) | 0.914 (0.018) |
| Fit time (s) | 2.4 (0.7) | 3.4 (0.8) |
| Marginal variance of $w$, interior cells [nominal 1] | 1.07 | – |
| Marginal variance of $w$, edge cells | 2.02 | – |

S3.2. *S0: accuracy of the field representation and of the Laplace engine.* We draw $w$ on the $40 \times 30$ domain from the continuous Matérn process with $\rho_w = 400$ km and $\sigma_w = 1$, and Bernoulli outcomes from $\mathrm{logit}\,\pi_{st} = \beta_0 + 0.5\,x_{st} + w(s)$ with $x_{st} \sim \mathcal{N}(0,1)$, in two settings with 50 replicates each. In the *common* setting, $\beta_0 = -2$ with 40 weeks (prevalence about 16%). In the *rare* setting, $\beta_0 = -4.5$ with 104 weeks (prevalence about 2%, as in the data). We also report the marginal variance of the lattice field itself, from 2,000 exact draws, in interior cells and at the edge of the domain. Table S2 summarises the results.

The lattice field has marginal variance 1.07 in cells at least eight cells (about 440 km) from the edge of the domain, against a nominal 1, and 2.02 at the edge. This is the doubling induced by the reflecting boundary that motivates the barrier model of Remark 1. Although the data are not generated from the lattice model, the range is recovered with little bias in both settings ($\hat{\rho}_w = 420$ and 409 km against 400). The standard deviation is underestimated by about 12% ($\hat{\sigma}_w = 0.88$ against 1). The bias is the same at prevalences of 16% and 2%, so it is not mainly due to the limited information in rare binary outcomes. A plausible cause is the mismatch between the lattice model and the continuous process, including the inflated variance near the boundary, since only about 28% of the cells are at least eight cells from the edge. We have not tested this explanation. The posterior mean of $w$ has correlation 0.95 with the truth in the common setting and 0.91 in the rare one. Plug-in 95% intervals cover $b_1$ in 88% of replicates in the common setting and 96% in the rare one (Monte Carlo SE about 4.6% and 2.8%). They cover $\beta_0 + w(s)$ for 94.8% and 94.3% of cells. Ignoring the uncertainty in $\boldsymbol{\theta}$ therefore costs little here, including at the prevalence of the data. This holds for the coefficient of a covariate that is independent in space and time and independent of $w$, which is the easiest case. S0 does not check spatially smooth covariates that are correlated with the field, and S1 shows that the plug-in standard error of $\delta$ is too small when a space-time field is fitted. A fit takes 2–4 seconds on one core.

S3.3. *S1: state dependence versus persistent heterogeneity.* Three data-generating processes on a smaller $16 \times 14$ domain (224 cells, 6 countries, 104 training weeks, 20 replicates each) produce autocorrelation in $Y_{st}$ for different reasons:

- *state dependence*: $\delta = 1$ and a weak field ($\sigma_w = 0.3$);
- *heterogeneity*: no state dependence ($\delta = 0$), a strong static field ($\sigma_w = 1.3$) and a space-time AR(1) field ($\sigma_v = 0.7$, $\phi = 0.9$ per week, $\rho_v = 300$ km);
- *mixed*: $\delta = 0.5$, $\sigma_w = 1$ and a space-time field with $\sigma_v = 0.5$ and $\phi = 0.9$.

Two further truths repeat the heterogeneity and mixed scenarios with a more persistent space-time field ($\phi = 0.97$ per week). We fit model (3.1) with lags only, with lags and $w$, and with lags, $w$ and $v$. The space-time field $v$ is resolved at weekly steps ($\Delta = 1$, the resolution of the data-generating process), at four-week steps ($\Delta = 4$) and at eight-week steps ($\Delta = 8$, the resolution of the regional state-dependence fits in the application). Table S3 shows the three main truths at $\Delta = 1$ and $\Delta = 4$. The $\Delta = 8$ fits, the two persistent truths and the fit times are reported in the text only.

Under state dependence, ignoring the latent field inflates $\hat{\delta}$ by about a quarter (1.26; coverage 0.60), and all latent specifications are approximately unbiased (0.98–1.03) with coverage of 0.90–0.95. Under heterogeneity the contrast is much sharper. With lags only, $\hat{\delta} = 1.61$ although the true value is zero. Adding the static field reduces the estimate to 0.46, but no interval covers zero: slowly varying episodes act as state dependence. Only the models with the space-time field bring the estimate close to zero: $\hat{\delta}$ is 0.05 with four-week steps and 0.01 with weekly steps, both within Monte Carlo error of zero (Monte Carlo SE about 0.035). The mixed scenario behaves in the same way. The static field leaves an upward bias (0.78 against 0.5), which the space-time field removes (0.55 and 0.54, Monte Carlo SE about 0.04).

The standard errors of $\hat{\delta}$ are too small when $v$ is fitted. In the heterogeneity and mixed scenarios the empirical SD exceeds the mean standard error by 27–43% (ratios 1.27–1.43), and coverage is 0.80–0.90 (Monte Carlo SE 0.07–0.09). In the state-dependence scenario, where the truth has no $v$, the ratio is 0.92–1.02 for the four latent models. With 20 replicates each ratio has a Monte Carlo SE of about 0.2, so the size of the shortfall is imprecise, but it appears in all twelve cells with $v$ in the truth and in the fitted model (ratios 1.19–1.66, including the $\phi = 0.97$ truths and $\Delta = 8$, with coverage 0.80–0.90). Plug-in intervals for $\delta$ from the space-time model are therefore too short, by a factor of roughly 1.2 to 1.7 in these designs. The ratio is not stable enough to justify a single correction factor. In the application, plug-in standard errors from models with a space-time field should be read as too small, and an estimate about two plug-in standard errors from zero should not be read as clearly different from zero.

Resolving $v$ at four-week steps instead of weekly steps changes $\hat{\delta}$ little in these scenarios, and it is four to six times faster (ratios of mean fit time 3.9 to 5.6). Paired by replicate, the four-week estimate is larger by 0.034 (SE 0.006) under heterogeneity and by 0.010 (SE 0.008) in the mixed scenario. The space-time field has an AR(1) coefficient of 0.9 per week, so it changes slowly relative to the four-week history window. This argument is weaker at eight-week steps, since $0.9^8 \approx 0.43$. With $\Delta = 8$ the estimate is still unbiased under state dependence (0.98, coverage 0.95), but under heterogeneity it is 0.10 (empirical SD 0.14, mean SE 0.12, coverage 0.80), about three Monte Carlo SE above zero. Paired by replicate, it exceeds the weekly estimate by 0.086 (SE 0.012). In the mixed scenario it is 0.57 against 0.5 (coverage 0.85; 0.031 above the weekly estimate, SE 0.011). Coarse steps therefore leave part of the episodes to be absorbed by the lag. The bias is smaller when the episodes are more persistent. With $\phi = 0.97$, $\hat{\delta}$ at $\Delta = 8$ is 0.04 under heterogeneity (coverage 0.90) and 0.50 in the mixed scenario (coverage 0.85), and it exceeds the weekly estimate by 0.050 (SE 0.007) and 0.024 (SE 0.006). In these slow truths the static field alone leaves less bias than before (0.38 and 0.71, against 0.46 and 0.78), and the four-week and weekly estimates are within 0.03 of the truth. A fit at $\Delta = 8$ takes 23–27 seconds, against 84–101 at $\Delta = 4$ and 350–474 at $\Delta = 1$. Faster space-time dynamics ($\phi < 0.9$) are not covered, and neither are the 13-week steps of the joint and space-time fits in the application.

## S4. Further application results.

TABLE S3
*S1: estimation of the state-dependence parameter δ (20 replicates per scenario).*

| Estimator | Mean $\hat{\delta}$ | Emp. SD | Mean SE | Coverage |
|---|---|---|---|---|
| *State dependence ($\delta = 1$)* | | | | |
| Lags only | 1.263 | 0.107 | 0.136 | 0.60 |
| Lags + $w$ | 1.032 | 0.131 | 0.139 | 0.90 |
| Lags + $w$ + $v$ (4-week steps) | 0.990 | 0.135 | 0.140 | 0.95 |
| Lags + $w$ + $v$ (weekly steps) | 0.997 | 0.143 | 0.140 | 0.95 |
| XGBoost: share of mean \|SHAP\| for lags | 0.108 | 0.034 | – | – |
| *Heterogeneity ($\delta = 0$)* | | | | |
| Lags only | 1.614 | 0.386 | 0.099 | 0.00 |
| Lags + $w$ | 0.464 | 0.093 | 0.111 | 0.00 |
| Lags + $w$ + $v$ (4-week steps) | 0.045 | 0.152 | 0.119 | 0.90 |
| Lags + $w$ + $v$ (weekly steps) | 0.011 | 0.162 | 0.120 | 0.90 |
| XGBoost: share of mean \|SHAP\| for lags | 0.094 | 0.038 | – | – |
| *Mixed ($\delta = 0.5$)* | | | | |
| Lags only | 1.659 | 0.351 | 0.106 | 0.00 |
| Lags + $w$ | 0.782 | 0.111 | 0.115 | 0.25 |
| Lags + $w$ + $v$ (4-week steps) | 0.545 | 0.152 | 0.120 | 0.85 |
| Lags + $w$ + $v$ (weekly steps) | 0.535 | 0.172 | 0.120 | 0.80 |
| XGBoost: share of mean \|SHAP\| for lags | 0.116 | 0.044 | – | – |

S4.1. *Reconstructing the cell-week table.* The table built by the original pipeline has 70,033,444 rows but only 34,671,180 distinct cell-weeks (Section 2.1). In one partition, 613,459 cell-weeks appear once, while others appear 2, 3, 8, 16, 32, 48 and up to 648 times. The national indicators and IPC regions are attached by spatial intersection and then merged on cell and year. A cell that touches several countries or IPC regions is therefore multiplied at every join, and each copy combines indicators from different units.

The duplication matters for more than the covariates. The four-week event counts of the original pipeline are computed row by row within each cell, after sorting by cell and week. When a cell-week has copies, the rows immediately before a given row are copies of the same week, so the "past" count contains the current week's events, which are the outcome. We recomputed the original counts exactly and compared them with the true counts over the four previous weeks, on 69.4 million rows of the duplicated table, over all years and without the coverage filter. Among rows with an event, counting each copy separately, 22% have an original count that includes the current week's events while the true past count is zero, for both outcomes. Used directly as a predictor of the outcome on these rows, without any model, the original count reaches an AUC-PR of 0.883 for conflict and 0.833 for unrest. The true count reaches 0.495 and 0.450. These values describe the duplicated table as a whole. They are not comparable with the test scores of fitted models. The leak arises only in cell-weeks that have copies, and never in the first copy of a cell-week, which is the one the repaired table keeps. Copies come from cells that intersect several countries or several IPC regions; we have not checked how many of them lie on international borders.

The repaired table has one row per cell-week (Section 2.1). After ACLED coverage and the biome exclusion it has 19,364,639 cell-weeks in 59,339 cells and 366 weeks. The test period has 4,787,224 cell-weeks in 46,031 cells with complete climate records.

S4.2. *Joint prediction when both outcomes are observed.* When both outcomes are observed in the same places, the joint model cannot be expected to forecast either one better than univariate models that use both histories as covariates (Section 4.2), and it does not.

*Prediction..* All predictive probabilities below are posterior means of $\mathrm{expit}(\eta)$ over the same 200 posterior draws for every model. The joint model does not forecast either outcome alone better. Its test log score for conflict is 0.1021, against 0.1016 for the univariate static model and 0.1017 for the univariate space-time model (AUC-PR 0.464, 0.470 and 0.470, at a conflict prevalence of 4.2%), and for unrest 0.0481 against 0.0481 and 0.0482. In the test period both events occur in 0.35% of the 260,520 cell-weeks, and conflict occurs in 25.1% of cell-weeks with unrest against 3.9% of those without. Even so, the joint model does not predict the two outcomes together better than the product of the univariate models. Its joint log score is 0.14993, against 0.14991 for the product of the univariate space-time models (difference $+0.00003$, SE $0.0005$ clustered by country) and 0.14978 for the product of the univariate static models ($+0.0002$, SE $0.0006$). Within the joint model, drawing the two outcomes jointly rather than independently improves the joint log score by $0.0002$ (SE $0.0001$), and conditioning on same-week unrest improves the log score of conflict from 0.1021 to 0.1019 (AUC-PR 0.464 to 0.466). These are small effects. The dependence between the outcomes in a cell-week is largely carried by the histories and the static effects, which the univariate models also contain, and the shared field adds little to it at the weekly scale. As in S2, prediction cannot decide between escalation and shared episodes. The joint fit took about 4.5 hours, against 8 minutes for the univariate static model and 43 minutes for the univariate space-time model.

For prediction, scored with the same Monte Carlo draws for every model on 1,426,776 test cell-weeks, the joint model with one shared field and the product of the univariate static models are equivalent. The joint log scores are 0.12168 and 0.12170 (difference $-0.00002$, SE $0.0006$ clustered by country), and the product of the univariate space-time models is worse (0.12257). For conflict alone the log scores are 0.0585 (joint), 0.0590 (univariate static) and 0.0589 (univariate space-time), and AUC-PR is 0.526, 0.524 and 0.522, at a prevalence of 2.4%. Drawing the outcomes jointly improves the joint log score by $0.00004$ (SE $0.00003$). Conditioning on same-week unrest barely changes the conflict forecast (log score 0.05849 against 0.05853). The joint fit took 6.4 hours with one shared field and 4.7 hours for the extended model (warm-started), against 9 minutes for the univariate static model and 41 minutes for the univariate space-time model.

*Where the joint model predicts better..* The average gain hides heterogeneity. To locate it without using the test period, we conditioned the AME models on 2018–2021 at the hyperparameters of the main fit, scored 2022, and ranked $10^\circ \times 10^\circ$ windows, moved in steps of $5^\circ$ and with at least 50 cells, by the total gain in joint log score of the joint model over the product of the univariate static models. We then evaluated the main fit on 2023–2024 in the windows selected on 2022. The best window on 2022 covers Afghanistan and its borders (60–70°E, 30–40°N; 200 cells), with a gain of 0.0173 per cell-week (SE 0.0117, clustered by country). On 2023–2024 the same window keeps a gain of 0.0148 (SE 0.0110), against an average over the region of 0.00002. Among the five windows selected on 2022, all keep a positive gain on 2023–2024 (0.007 to 0.016), but only the window over northern Algeria and Tunisia is precise (0.0103, SE 0.0018). By country, Afghanistan (0.029 per cell-week) and Algeria (0.0035) have the largest total gains in both periods, followed in 2023–2024 by Nigeria (0.0057), Burkina Faso (0.0090) and Kenya (0.0045). In every selected window, the part of the gain due to the dependence between the outcomes is about zero ($|\cdot| \leq 0.0003$). The joint model predicts better there because the unrest data sharpen the latent fields that also drive conflict, and vice versa, not because it predicts co-occurrence better. In Afghanistan, for example, the change of regime in August 2021 shifted both outcomes together, a shared episode that a univariate model must learn from one outcome alone. The joint model also predicts worse in some places, most of all in Yemen ($-0.027$ per cell-week), where both events occur together in 6% of cell-weeks, and in Sudan ($-0.005$), where war broke out in April

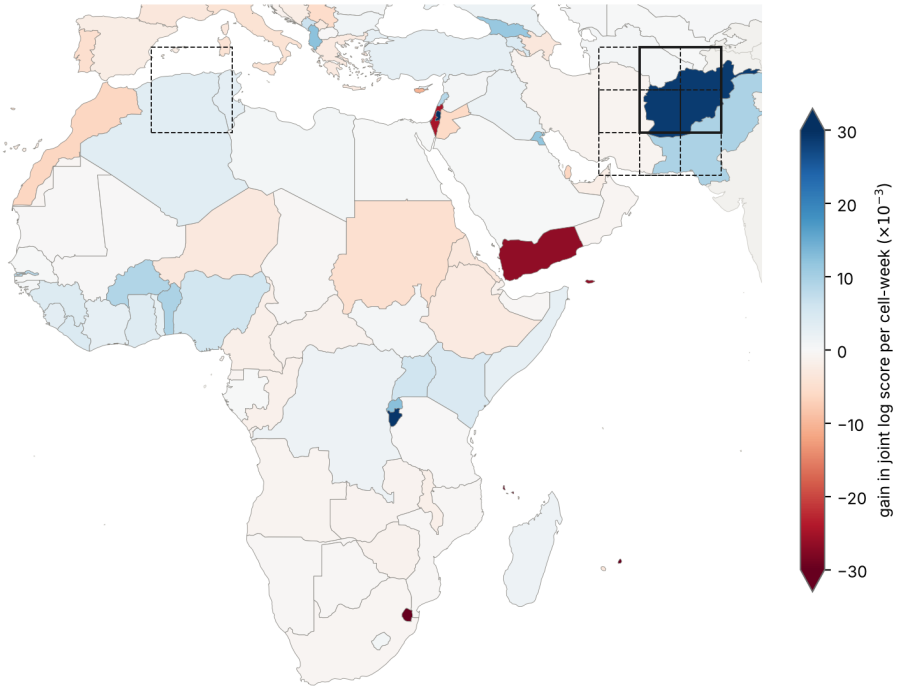



FIG S1. *Gain in joint log score per cell-week of the joint model over the product of the univariate static models, by country, on the 2023–2024 test period (blue: the joint model predicts better). Boxes: the five* $10^\circ \times 10^\circ$ *windows with the largest total gain on the 2022 validation year, selected without the test period; solid: the first.*

TABLE S4
*Latent blocks of the latent Gaussian model of Section 3.1 of the main paper, their precision matrices and hyperparameters.*

| Block | Dimension | Precision | Hyperparameters |
|---|---|---|---|
| fixed effects $\boldsymbol{\beta}, \delta, \boldsymbol{\alpha}$ | $p$ | $10^{-4}\boldsymbol{I}$ | – |
| country effects $\boldsymbol{u}$ | $n_c$ | $\sigma_u^{-2}\boldsymbol{I}$ | $\sigma_u$ |
| spatial field $\boldsymbol{w}$ | $n$ | $\boldsymbol{Q}_s(\rho_w, \sigma_w)$, SPDE precision | $\rho_w, \sigma_w$ |
| cell effects $\boldsymbol{e}$ | $n$ | $\sigma_e^{-2}\boldsymbol{I}$ | $\sigma_e$ |
| space-time field $\boldsymbol{v}$ | $nT_v$ | $\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$ | $\rho_v, \sigma_v, \phi$ |

2023 (Figure S1). These losses offset the gains, so over the whole region the two models predict equally well. A latent field shared by the two outcomes helps where the 2018–2022 relation between them persists, and hurts where it breaks.

## S5. Additional tables.

## REFERENCES

BAKKA, H., VANHATALO, J., ILLIAN, J. B., SIMPSON, D. and RUE, H. (2019). Non-stationary Gaussian models with physical barriers. *Spatial Statistics* **29** 268–288.

TABLE S5

*Boundary treatments on a $60 \times 30$ lattice of $0.5^\circ$ cells with a sea strait of 2, 4 or 6 cells (about 110, 220, 330 km): marginal variance of a coastal and of an inland land cell (nominal 1) and correlation between the two coastal cells facing each other across the strait, from 4,000 exact draws with $\rho = 400$ km. "Land only" removes the strait from the lattice; "sea = land" keeps it as ordinary cells without data. The strait lies inside a closed rectangle. This is not the geometry of the application, which has an open coast, a buffer of sea cells and a reflecting outer edge, and the fitted ranges there are both above and below 400 km, so the table shows the behaviour of the barrier only qualitatively.*

| Strait | | Land only | Sea = land | $f = 0.1$ | $f = 0.2$ | $f = 0.5$ |
|---|---|---|---|---|---|---|
| 2 cells | var. coast | 1.98 | 1.07 | 1.80 | 1.67 | 1.29 |
| | var. inland | 1.10 | 1.06 | 1.09 | 1.11 | 1.09 |
| | corr. across | -0.01 | 0.49 | -0.03 | 0.01 | 0.16 |
| 4 cells | var. coast | 2.00 | 1.08 | 1.77 | 1.69 | 1.29 |
| | var. inland | 1.06 | 1.05 | 1.09 | 1.09 | 1.09 |
| | corr. across | 0.00 | 0.28 | 0.01 | 0.01 | 0.04 |
| 6 cells | var. coast | 2.02 | 1.04 | 1.76 | 1.66 | 1.26 |
| | var. inland | 1.06 | 1.05 | 1.08 | 1.07 | 1.07 |
| | corr. across | -0.02 | 0.14 | 0.01 | 0.01 | 0.01 |

TABLE S6

*S2: full results for the escalation effect $\kappa_{\mathrm{U}\to\mathrm{C}}$ of recent unrest on conflict, by scenario and estimator, over 20 replicates per scenario. Mean $\hat{\kappa}_{\mathrm{U}\to\mathrm{C}}$, its empirical SD across replicates and the mean of the estimated SEs; coverage of the nominal 95% intervals for $\kappa_{\mathrm{U}\to\mathrm{C}}$ with its Monte Carlo SE; proportion of replicates whose interval excludes zero; mean estimate of the own-lag coefficient $d_C$ (truth 0.3); mean and SD of the estimated loading $\lambda$ of the shared space-time field (truth 0.8, joint model only); and mean log score, AUC-PR and onset AUC-PR of the conflict forecasts over the 26 test weeks. Onset rows are those of cells without a conflict event in the previous 52 weeks. The last two scenarios add unrest-only episodes (AR(1) with coefficient 0.95, Matérn innovations of range 200 km, marginal SD 0.6), which the joint model with one shared field must absorb into that field.*

| Estimator | Mean $\hat{\kappa}$ | Emp. SD | Mean SE | Coverage (MC SE) | Reject $\kappa = 0$ | Mean $\hat{d}_c$ | Mean $\hat{\lambda}$ (SD) | Log score | AUC-PR | Onset AUC-PR |
|---|---|---|---|---|---|---|---|---|---|---|
| *No escalation, $\kappa = 0$* | | | | | | | | | | |
| Logistic regression | 0.695 | 0.193 | 0.129 | 0.05 (0.05) | 0.95 | 1.54 | – | 0.1372 | 0.063 | 0.022 |
| Static field $w_c$ | 0.426 | 0.161 | 0.135 | 0.15 (0.08) | 0.85 | 0.82 | – | 0.1334 | 0.090 | 0.036 |
| Static field $w_c$ + space-time field | 0.269 | 0.158 | 0.140 | 0.45 (0.11) | 0.55 | 0.38 | – | 0.1336 | 0.090 | 0.036 |
| Joint model, shared space-time field | 0.042 | 0.166 | 0.138 | 0.90 (0.07) | 0.10 | 0.38 | 0.83 (0.14) | 0.1338 | 0.088 | 0.034 |
| Joint model, shared + unrest-specific space-time fields | 0.058 | 0.165 | 0.139 | 0.85 (0.08) | 0.15 | 0.36 | 0.77 (0.14) | 0.1336 | 0.089 | 0.035 |
| *Escalation, $\kappa = 0.4$* | | | | | | | | | | |
| Logistic regression | 1.026 | 0.163 | 0.116 | 0.00 (0.00) | 1.00 | 1.52 | – | 0.1430 | 0.077 | 0.025 |
| Static field $w_c$ | 0.794 | 0.128 | 0.122 | 0.10 (0.07) | 1.00 | 0.80 | – | 0.1392 | 0.105 | 0.045 |
| Static field $w_c$ + space-time field | 0.659 | 0.108 | 0.127 | 0.50 (0.11) | 1.00 | 0.37 | – | 0.1393 | 0.105 | 0.045 |
| Joint model, shared space-time field | 0.434 | 0.111 | 0.126 | 0.95 (0.05) | 0.95 | 0.36 | 0.83 (0.15) | 0.1396 | 0.103 | 0.044 |
| Joint model, shared + unrest-specific space-time fields | 0.450 | 0.109 | 0.126 | 0.95 (0.05) | 0.95 | 0.35 | 0.78 (0.14) | 0.1394 | 0.104 | 0.045 |
| *No escalation, $\kappa = 0$, with unrest-only episodes* | | | | | | | | | | |
| Logistic regression | 0.683 | 0.201 | 0.121 | 0.10 (0.07) | 0.90 | 1.53 | – | 0.1373 | 0.063 | 0.025 |
| Static field $w_c$ | 0.411 | 0.164 | 0.128 | 0.20 (0.09) | 0.80 | 0.82 | – | 0.1335 | 0.089 | 0.036 |
| Static field $w_c$ + space-time field | 0.281 | 0.158 | 0.135 | 0.45 (0.11) | 0.55 | 0.38 | – | 0.1337 | 0.090 | 0.036 |
| Joint model, shared space-time field | −0.012 | 0.166 | 0.131 | 0.85 (0.08) | 0.15 | 0.47 | 1.19 (0.32) | 0.1338 | 0.088 | 0.039 |
| Joint model, shared + unrest-specific space-time fields | 0.064 | 0.169 | 0.133 | 0.85 (0.08) | 0.15 | 0.37 | 0.81 (0.20) | 0.1338 | 0.089 | 0.037 |
| *Escalation, $\kappa = 0.4$, with unrest-only episodes* | | | | | | | | | | |
| Logistic regression | 1.017 | 0.175 | 0.108 | 0.00 (0.00) | 1.00 | 1.51 | – | 0.1435 | 0.079 | 0.031 |
| Static field $w_c$ | 0.786 | 0.140 | 0.115 | 0.15 (0.08) | 1.00 | 0.80 | – | 0.1396 | 0.107 | 0.048 |
| Static field $w_c$ + space-time field | 0.679 | 0.136 | 0.122 | 0.35 (0.11) | 1.00 | 0.37 | – | 0.1398 | 0.107 | 0.047 |
| Joint model, shared space-time field | 0.385 | 0.125 | 0.118 | 0.90 (0.07) | 0.90 | 0.44 | 1.17 (0.29) | 0.1401 | 0.105 | 0.045 |
| Joint model, shared + unrest-specific space-time fields | 0.467 | 0.137 | 0.120 | 0.85 (0.08) | 1.00 | 0.35 | 0.81 (0.19) | 0.1399 | 0.106 | 0.045 |

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TABLE S7

*Joint model of conflict and unrest, by region (Section 5 of the main paper): all hyperparameters and lag coefficients (posterior mean, with SE in parentheses) of the two univariate conflict models and of the joint model, fitted on 2018–2022. Subscripts C and U denote conflict and unrest. In the univariate model with $u + e + w + v$, the space-time field $v$ belongs to conflict alone; in the joint model it is shared, with loading $\lambda$ for unrest. Histories are $\log(1 +$ events in the previous four weeks$)$; neighbour histories are means over the rook neighbours, standardised on the training rows. The log marginal likelihood of the joint model includes the unrest data and is not comparable with those of the univariate models. Test scores are for 2023–2024 in the same region. The field $v$ is defined on the observed $1^\circ$ cells only, with reflecting edges, which departs from the boundary rule of Remark 1 of the main paper. SEs are plug-in values; in S1 and S2 such SEs are too small when a space-time field is fitted (ratio of empirical SD to mean SE up to 1.66 in S1 and 1.20 in S2 without escalation), so intervals from these models are likely too narrow. With a single shared space-time field, unrest-only episodes are absorbed into $v$, which can bias $\kappa_{\mathrm{U}\to\mathrm{C}}$ towards zero; in S2 they inflated $\hat{\lambda}$ instead. The extended model has an additional unrest-specific space-time field $v_{\mathrm{U}}$. Test scores here use the probit approximation to $E\{\mathrm{expit}(\eta)\}$; the model comparisons in the text use common Monte Carlo draws.*

| Quantity | Conflict only $u + e + w$ | Conflict only $u + e + w + v$ | Joint shared $v$ | Joint shared $v$ + $v_{\mathrm{U}}$ |
|---|---|---|---|---|
| *West Africa and Sahel* | | | | |
| 2,505 cells, 22 countries, 906,810 rows per outcome | | | | |
| $v$ on 641 observed $1^\circ$ cells (reflecting edges), 28 steps of 13 weeks | | | | |
| Hyperparameters | | | | |
| $\sigma_{u,\mathrm{C}}$ | 0.69 | 0.67 | 0.69 | 0.67 |
| $\sigma_{e,\mathrm{C}}$ | 0.65 | 0.64 | 0.66 | 0.65 |
| $\rho_{w,\mathrm{C}}$ (km) | 514 | 508 | 453 | 467 |
| $\sigma_{w,\mathrm{C}}$ | 1.70 | 1.74 | 1.39 | 1.44 |
| $\sigma_{u,\mathrm{U}}$ | – | – | 0.53 | 0.51 |
| $\sigma_{e,\mathrm{U}}$ | – | – | 1.34 | 1.36 |
| $\rho_{w,\mathrm{U}}$ (km) | – | – | 369 | 497 |
| $\sigma_{w,\mathrm{U}}$ | – | – | 0.21 | 0.34 |
| $\rho_v$ (km) | – | 423 | 499 | 445 |
| $\sigma_v$ | – | 0.69 | 1.01 | 0.99 |
| $\phi$ (per step) | – | 0.905 | 0.952 | 0.957 |
| $\lambda$ | – | – | 0.868 | 0.580 |
| $\rho_{v,\mathrm{U}}$ (km) | – | – | – | 787 |
| $\sigma_{v,\mathrm{U}}$ | – | – | – | 0.49 |
| $\phi_{\mathrm{U}}$ (per step) | – | – | – | 0.687 |
| Conflict equation | | | | |
| $\delta_{\mathrm{C}}$: own history | 0.575 (0.016) | 0.444 (0.017) | 0.459 (0.017) | 0.448 (0.017) |
| $\kappa$: unrest history | 0.138 (0.031) | 0.131 (0.032) | 0.044 (0.032) | 0.095 (0.032) |
| $\delta_{\mathrm{C}}^{\mathrm{nb}}$: neighbours' conflict | 0.129 (0.007) | 0.051 (0.008) | 0.061 (0.008) | 0.054 (0.008) |
| $\kappa^{\mathrm{nb}}$: neighbours' unrest | 0.005 (0.008) | 0.003 (0.008) | −0.020 (0.008) | −0.005 (0.008) |
| Unrest equation | | | | |
| $\delta_{\mathrm{U}}$: own history | – | – | 0.351 (0.029) | 0.269 (0.029) |
| $\kappa_{\mathrm{U}}$: conflict history | – | – | 0.066 (0.031) | 0.111 (0.031) |
| $\delta_{\mathrm{U}}^{\mathrm{nb}}$: neighbours' unrest | – | – | 0.017 (0.009) | −0.004 (0.009) |
| $\kappa_{\mathrm{U}}^{\mathrm{nb}}$: neighbours' conflict | – | – | −0.042 (0.013) | −0.008 (0.014) |
| Fit | | | | |
| Log marginal likelihood | −49530.1 | −49080.6 | −76591.8 | −76461.0 |
| Fit time (h) | 0.1 | 0.7 | 4.5 | 8.2 |
| Test AUC-PR, conflict | 0.470 | 0.471 | 0.466 | 0.470 |
| Test log score, conflict | 0.1020 | 0.1026 | 0.1031 | 0.1029 |
| Test AUC-PR, unrest | – | – | 0.289 | 0.283 |
| Test log score, unrest | – | – | 0.0490 | 0.0496 |
| *AME region* | | | | |
| 13,719 cells, 93 countries, 4,921,758 rows per outcome | | | | |
| $v$ on 3,637 observed $1^\circ$ cells (reflecting edges), 14 steps of 26 weeks | | | | |
| Hyperparameters | | | | |
| $\sigma_{u,\mathrm{C}}$ | 0.50 | 0.50 | 0.53 | 0.52 |
| $\sigma_{e,\mathrm{C}}$ | 0.96 | 0.97 | 0.99 | 0.99 |
| $\rho_{w,\mathrm{C}}$ (km) | 562 | 583 | 632 | 598 |

TABLE S8

*S1: estimation of the state-dependence parameter $\delta$ for all five truths and all step lengths of the space-time field $v$ (20 replicates per truth). The two truths with slow $v$ have an AR(1) coefficient of 0.97 per week instead of 0.9. SD/SE is the ratio of the empirical SD of $\hat{\delta}$ to the mean estimated SE; values above 1 mean that the plug-in intervals are too narrow. Time is the mean fit time per replicate.*

| Estimator | Mean $\hat{\delta}$ | Emp. SD | Mean SE | SD/SE | Coverage | Time (s) |
|---|---|---|---|---|---|---|
| *State dependence ($\delta = 1$)* | | | | | | |
| Lags only | 1.263 | 0.107 | 0.136 | 0.78 | 0.60 | 0.1 |
| Lags + $w$ | 1.032 | 0.131 | 0.139 | 0.94 | 0.90 | 0.9 |
| Lags + $w$ + $v$ (8-week steps) | 0.981 | 0.128 | 0.140 | 0.92 | 0.95 | 27.3 |
| Lags + $w$ + $v$ (4-week steps) | 0.990 | 0.135 | 0.140 | 0.96 | 0.95 | 101.5 |
| Lags + $w$ + $v$ (weekly steps) | 0.997 | 0.143 | 0.140 | 1.02 | 0.95 | 449.6 |
| *Heterogeneity ($\delta = 0$)* | | | | | | |
| Lags only | 1.614 | 0.386 | 0.099 | 3.88 | 0.00 | 0.1 |
| Lags + $w$ | 0.464 | 0.093 | 0.111 | 0.84 | 0.00 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.097 | 0.140 | 0.117 | 1.19 | 0.80 | 26.4 |
| Lags + $w$ + $v$ (4-week steps) | 0.045 | 0.152 | 0.119 | 1.28 | 0.90 | 95.1 |
| Lags + $w$ + $v$ (weekly steps) | 0.011 | 0.162 | 0.120 | 1.35 | 0.90 | 474.2 |
| *Mixed ($\delta = 0.5$)* | | | | | | |
| Lags only | 1.659 | 0.351 | 0.106 | 3.32 | 0.00 | 0.1 |
| Lags + $w$ | 0.782 | 0.111 | 0.115 | 0.96 | 0.25 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.567 | 0.144 | 0.119 | 1.21 | 0.85 | 24.6 |
| Lags + $w$ + $v$ (4-week steps) | 0.545 | 0.152 | 0.120 | 1.27 | 0.85 | 90.0 |
| Lags + $w$ + $v$ (weekly steps) | 0.535 | 0.172 | 0.120 | 1.43 | 0.80 | 349.7 |
| *Heterogeneity, slow $v$ ($\delta = 0$, $\phi = 0.97$ per week)* | | | | | | |
| Lags only | 1.649 | 0.418 | 0.100 | 4.18 | 0.00 | 0.1 |
| Lags + $w$ | 0.384 | 0.115 | 0.113 | 1.01 | 0.05 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.037 | 0.160 | 0.119 | 1.35 | 0.90 | 25.7 |
| Lags + $w$ + $v$ (4-week steps) | 0.006 | 0.170 | 0.119 | 1.42 | 0.95 | 91.8 |
| Lags + $w$ + $v$ (weekly steps) | -0.013 | 0.165 | 0.120 | 1.37 | 0.90 | 464.9 |
| *Mixed, slow $v$ ($\delta = 0.5$, $\phi = 0.97$ per week)* | | | | | | |
| Lags only | 1.650 | 0.367 | 0.107 | 3.42 | 0.00 | 0.1 |
| Lags + $w$ | 0.711 | 0.144 | 0.118 | 1.22 | 0.55 | 1.0 |
| Lags + $w$ + $v$ (8-week steps) | 0.503 | 0.194 | 0.121 | 1.60 | 0.85 | 22.9 |
| Lags + $w$ + $v$ (4-week steps) | 0.491 | 0.198 | 0.122 | 1.63 | 0.85 | 84.0 |
| Lags + $w$ + $v$ (weekly steps) | 0.479 | 0.203 | 0.122 | 1.66 | 0.85 | 468.4 |


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